STM32F411 IMU - LAB1

Illustrated guide: hardware, configuration and dataset acquisition.

This English guide contains 14 selected screenshots from 151 source PNGs and one user-supplied hardware photo. Original evidence, the selection table and full README are included.

Checked configuration: STM32F411CEU6 + MPU6500; PB8 SCL/PB9 SDA; SYSCLK 84 MHz; USB 48 MHz; TIM2 100 Hz; TIM5 1 us; I2C DMA and USB CDC on COM5.

Recorded results: 21-byte frames; CRC-8 polynomial 0x07; 10 s benchmark with 1000 frames, 0 CRC errors, 0 lost frames, 99.96 Hz. Dataset: 15 CSV files, 5 classes x 3 sessions.

README.md provides PowerShell commands, directory structure, reproduction steps and troubleshooting. collect.py refuses to overwrite existing class/session files. Back up or use a new directory before collecting again.

The hardware photo shows the assembly but does not verify individual pins or voltage. No new hardware test was performed for this documentation update. Assignment substitutions and evidence gaps remain documented.

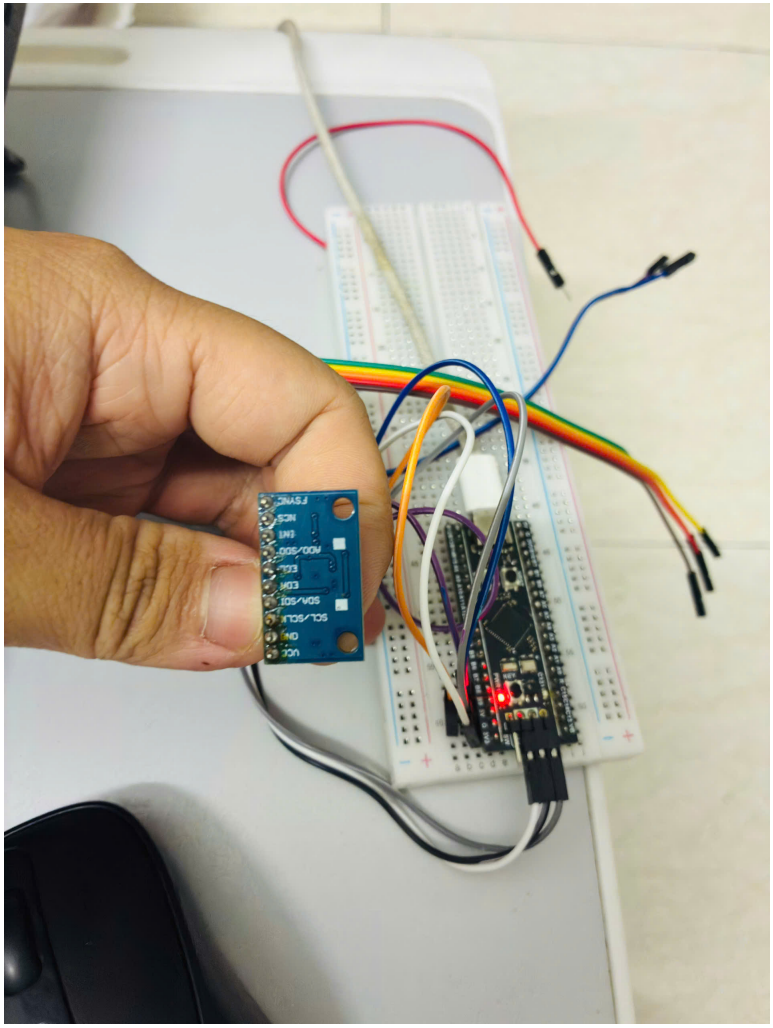
Presentation images are direct crops without regenerated text or numbers. Captions identify the main software. Original evidence and SHA-256 hashes are unchanged. Application text inside screenshots is preserved in its original language.

English documentation: README.md, report_EN.md, docs/EVIDENCE_INDEX_EN.md, docs/CSV_CHECK_EN.md and docs/VERIFICATION_EN.md. Vietnamese guide: README_VI.md and docs/LAB1_Huong_dan.pdf.

Actual hardware assembly

Actual hardware photo (no software)

The user-supplied photo shows an IMU module, STM32 board, breadboard and jumper wires. It does not verify individual pins, supply voltage or wire colors. PB8=SCL and PB9=SDA come from source configuration, not visual wire tracing.

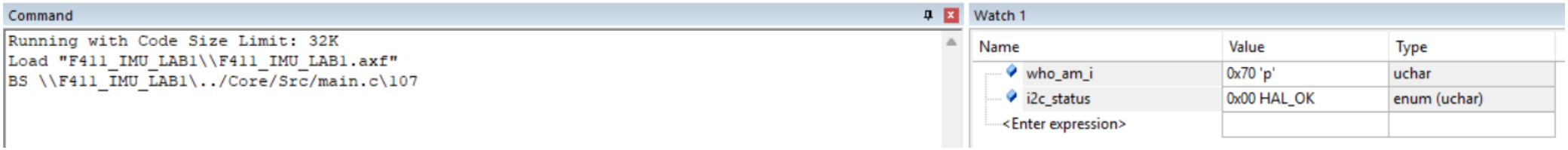


Original evidence: docs/evidence/00_hardware_original.jpg

01. Identify the MPU6500

Keil µVision

Watch: WHO_AM_I = 0x70 and I2C HAL_OK. This is the initial identification test.

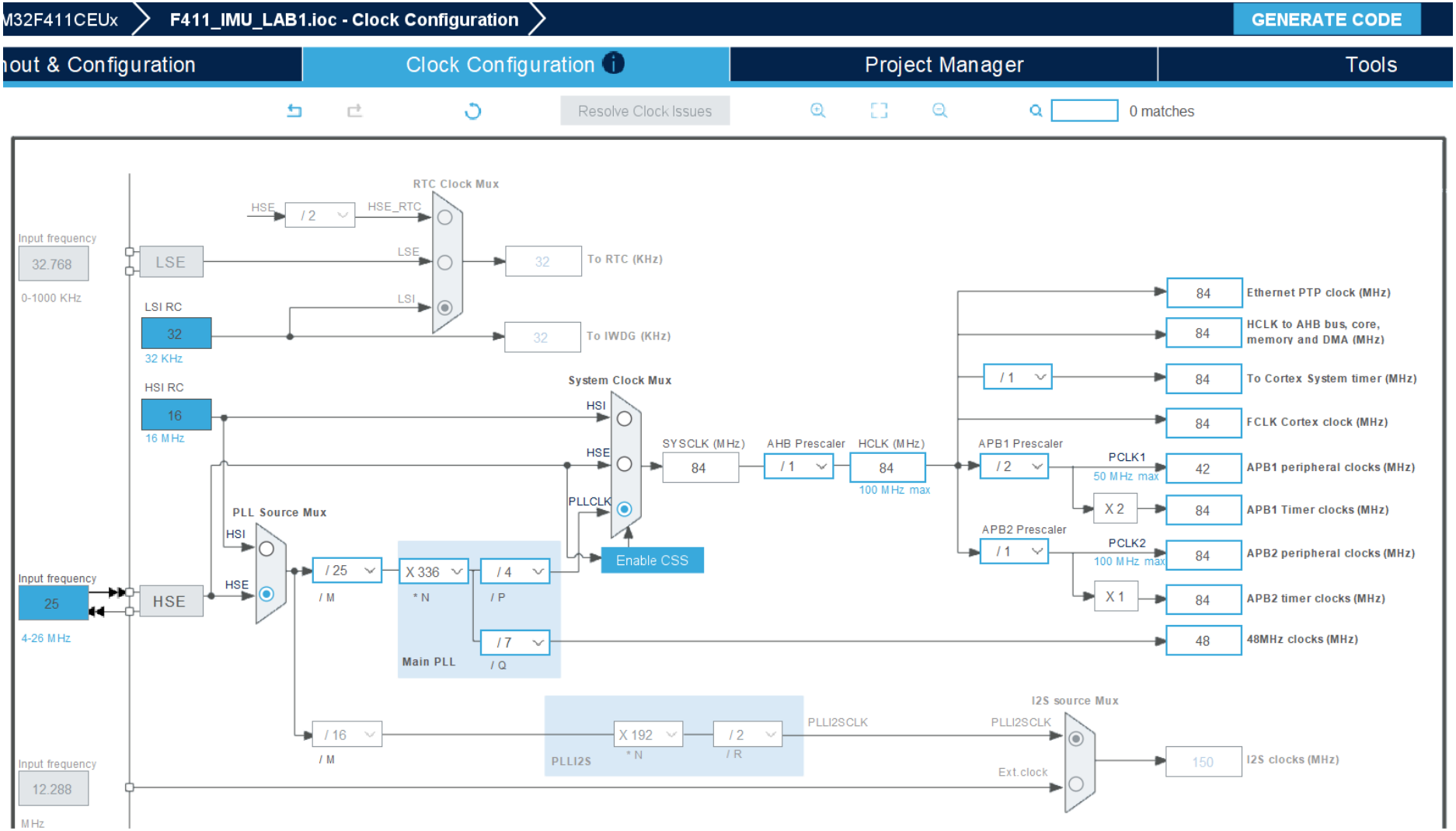


Original evidence: docs/evidence/14_whoami_original.png

02. SYSCLK 84 MHz and USB 48 MHz

STM32CubeMX

HSE 25 MHz; PLL M=25, N=336, P=4, Q=7. SYSCLK 84 MHz; USB 48 MHz.

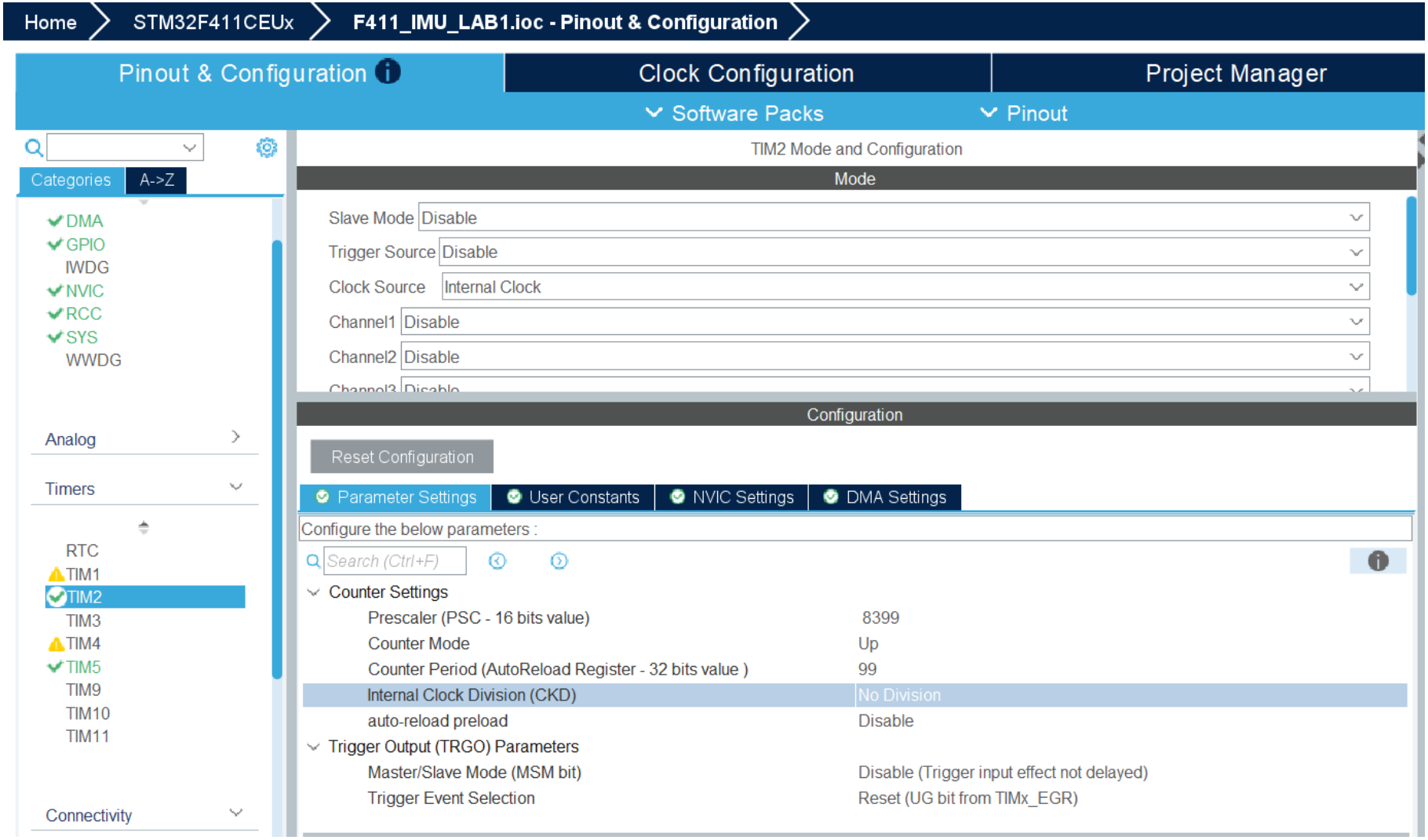


Original evidence: docs/evidence/01_clock_original.png

03. TIM2 sampling at 100 Hz

STM32CubeMX

PSC=8399; ARR=99; timer clock 84 MHz. 84 MHz / 8400 / 100 = 100 Hz.

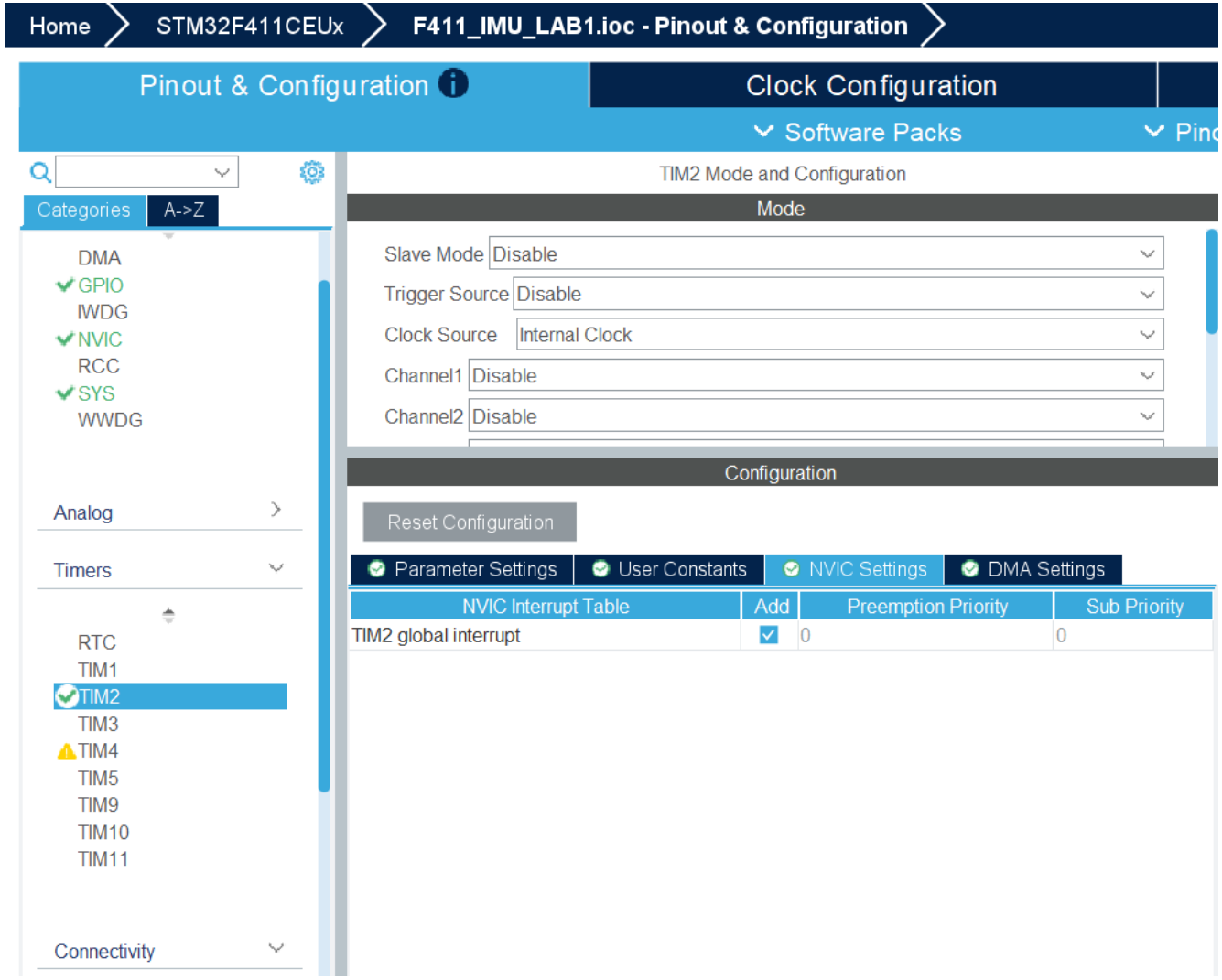


Original evidence: docs/evidence/02_tim2_original.png

04. TIM2 interrupt

STM32CubeMX

Enable TIM2 global interrupt. This image demonstrates interrupt enable only; final timer values appear in the previous image.

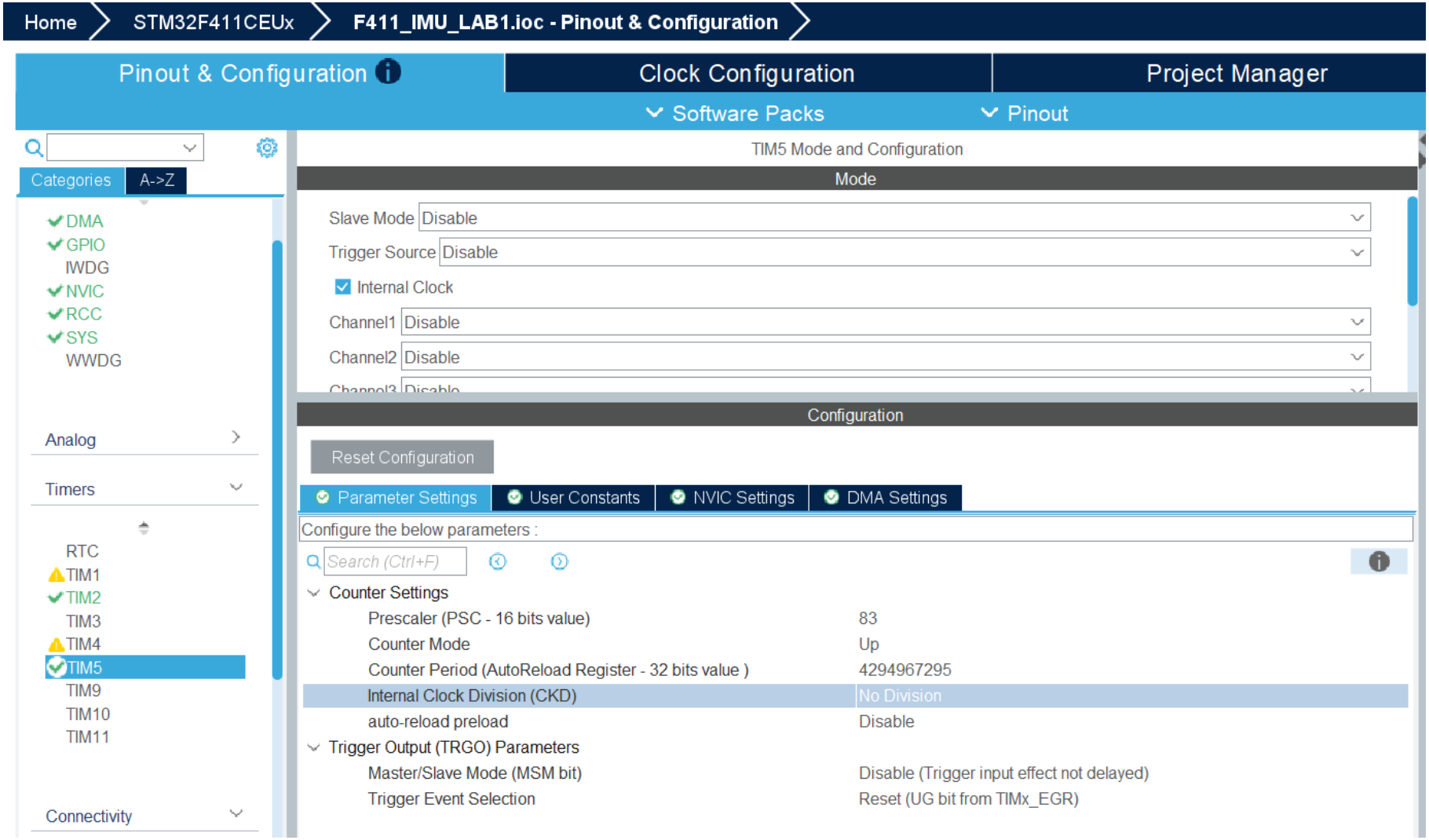


Original evidence: docs/evidence/12_tim2_irq_original.png

05. TIM5 timestamps at 1 us

STM32CubeMX

PSC=83; Period=4294967295 (0xFFFFFFFF). Timer clock 84 MHz gives 1 tick = 1 us.

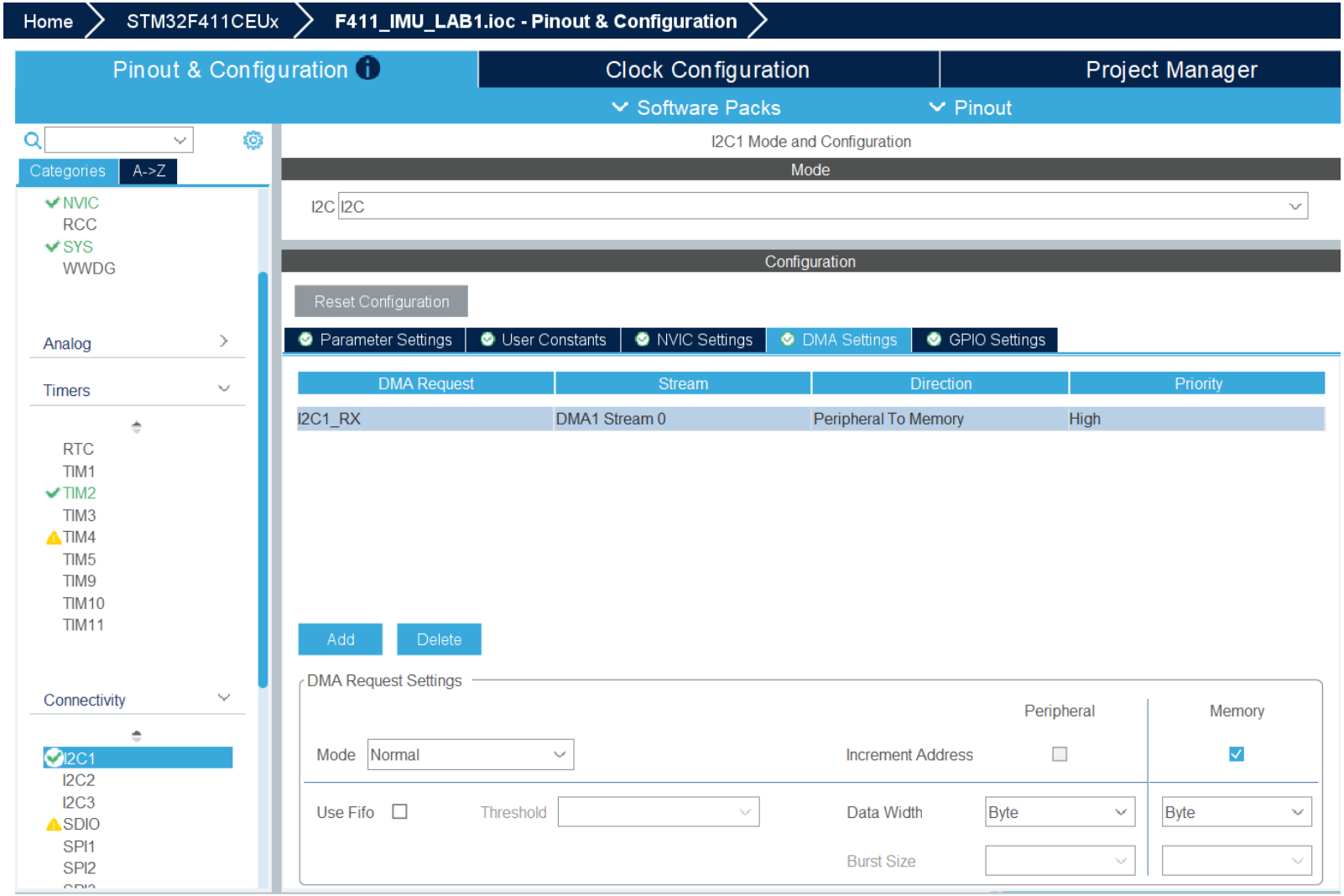


Original evidence: docs/evidence/03_tim5_original.png

06. I2C1 RX DMA

STM32CubeMX

DMA1 Stream0; Peripheral To Memory; High; Normal; Memory Increment; Byte data. Channel 1 is checked against MSP source.

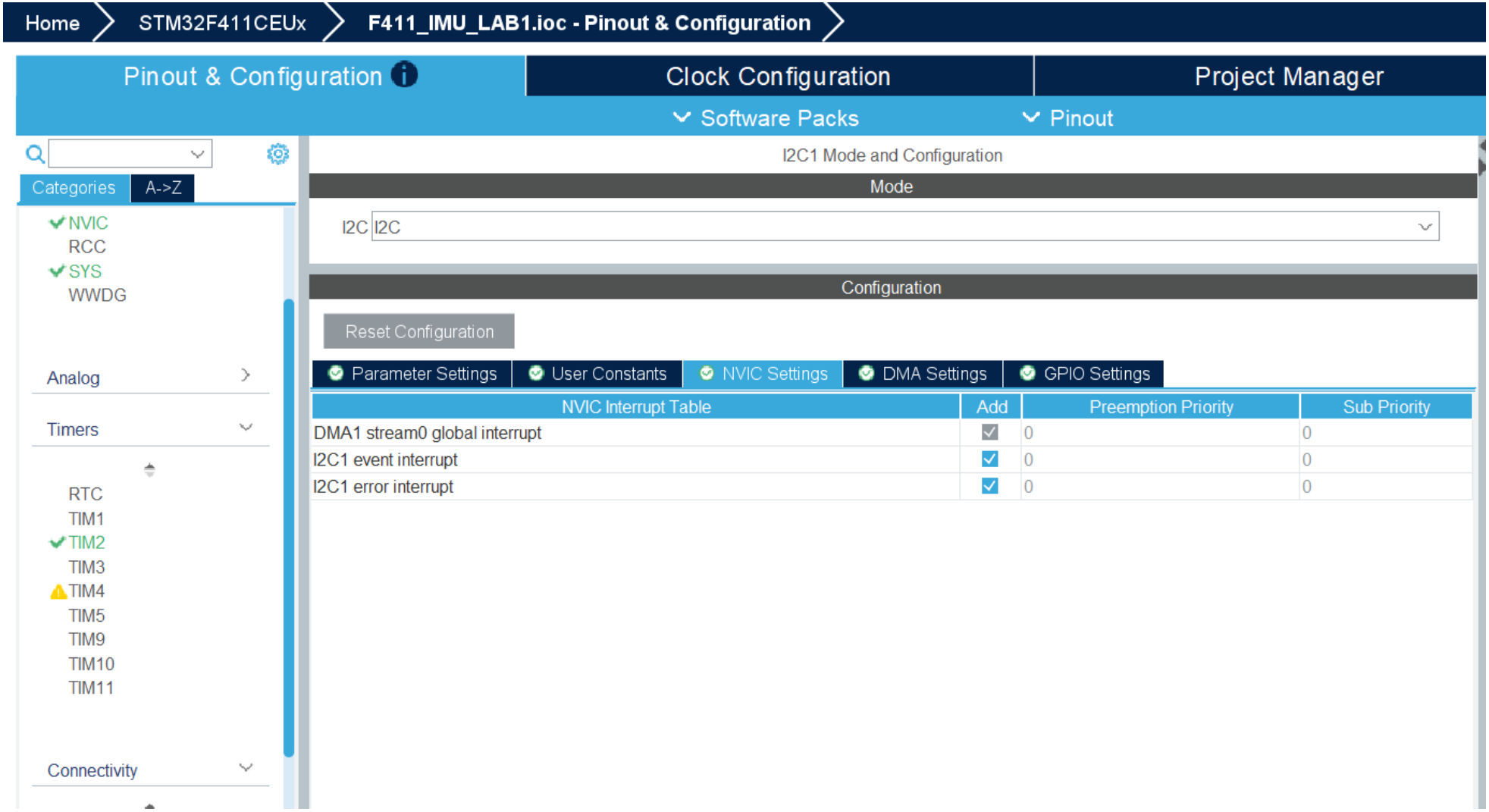


Original evidence: docs/evidence/04_dma_original.png

07. NVIC for I2C and DMA

STM32CubeMX

DMA1 Stream0 global interrupt, I2C1 event and I2C1 error interrupts are all enabled.

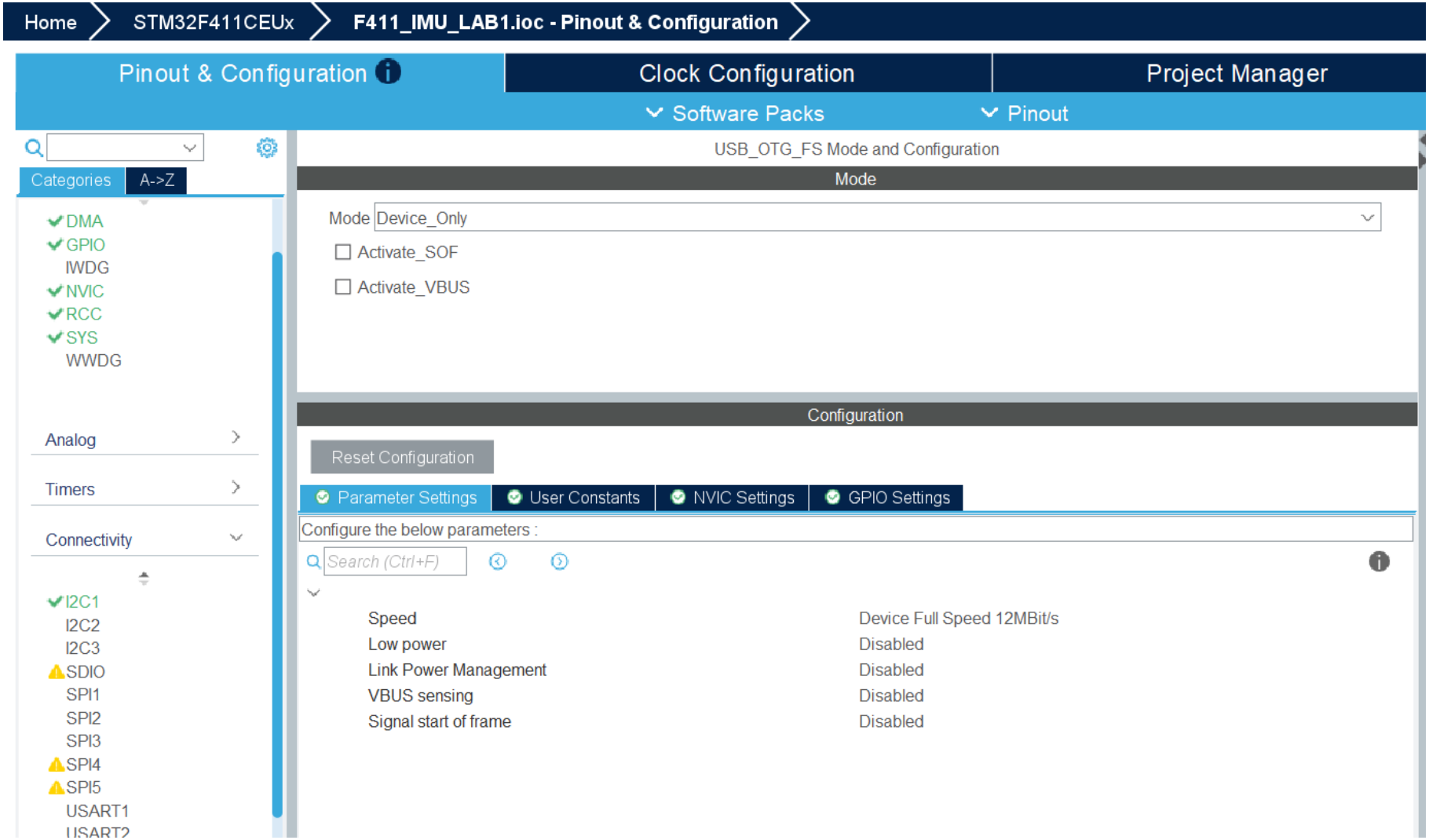


Original evidence: docs/evidence/05_nvic_original.png

08. USB OTG FS Device Only

STM32CubeMX

USB_OTG_FS: Device_Only; Device Full Speed 12 MBit/s. This is USB bus speed, not the sampling rate.

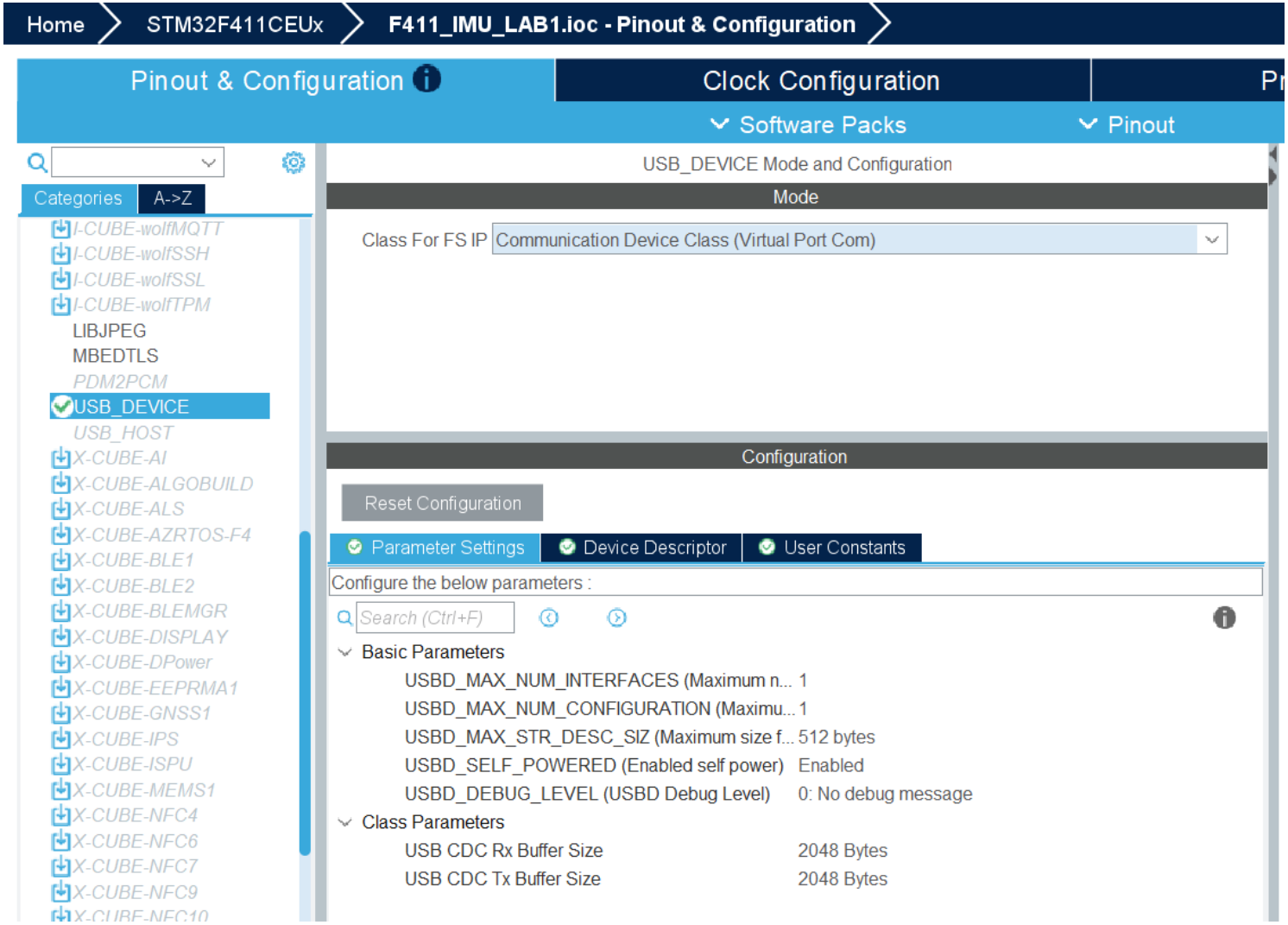


Original evidence: docs/evidence/13_usb_mode_original.png

09. USB_DEVICE CDC

STM32CubeMX

Class For FS IP: Communication Device Class (Virtual Port Com). This creates a virtual COM port over USB.



Original evidence: docs/evidence/06_cdc_original.png

10. Keil build and Flash Verify OK

Keil µVision

Build: 0 Error(s), 0 Warning(s). Programming Done and Verify OK. These are recorded results, not a new build or flash.

Build Output

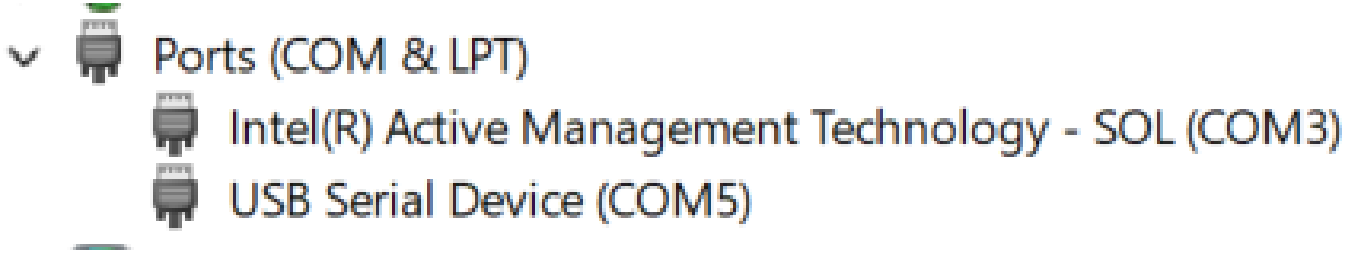
```
Build started: Project: F411_IMU_LAB1
*** Using Compiler 'V6.24', folder: 'C:\Users\LENOVO\AppData\Local\Keil_v5\ARM\ARMCLANG\Bin'
Build target 'F411_IMU_LAB1'
compiling main.c...
linking...
Program Size: Code=30102 RO-data=474 RW-data=248 ZI-data=9208
FromELF: creating hex file...
"F411_IMU_LAB1\F411_IMU_LAB1.axf" - 0 Error(s), 0 Warning(s).
Build Time Elapsed:  00:00:02
Load "F411_IMU_LAB1\\F411_IMU_LAB1.axf"
Erase Done.
Programming Done.
Verify OK.
Flash Load finished at 14:52:48
Load "F411_IMU_LAB1\\F411_IMU_LAB1.axf"
Erase Done.
Programming Done.
Verify OK.
Flash Load finished at 14:54:10
```

Original evidence: docs/evidence/07_build_flash_original.png

11. Windows detects COM5

Windows Device Manager

Device Manager shows USB Serial Device (COM5). COM3 in the same list belongs to another device.

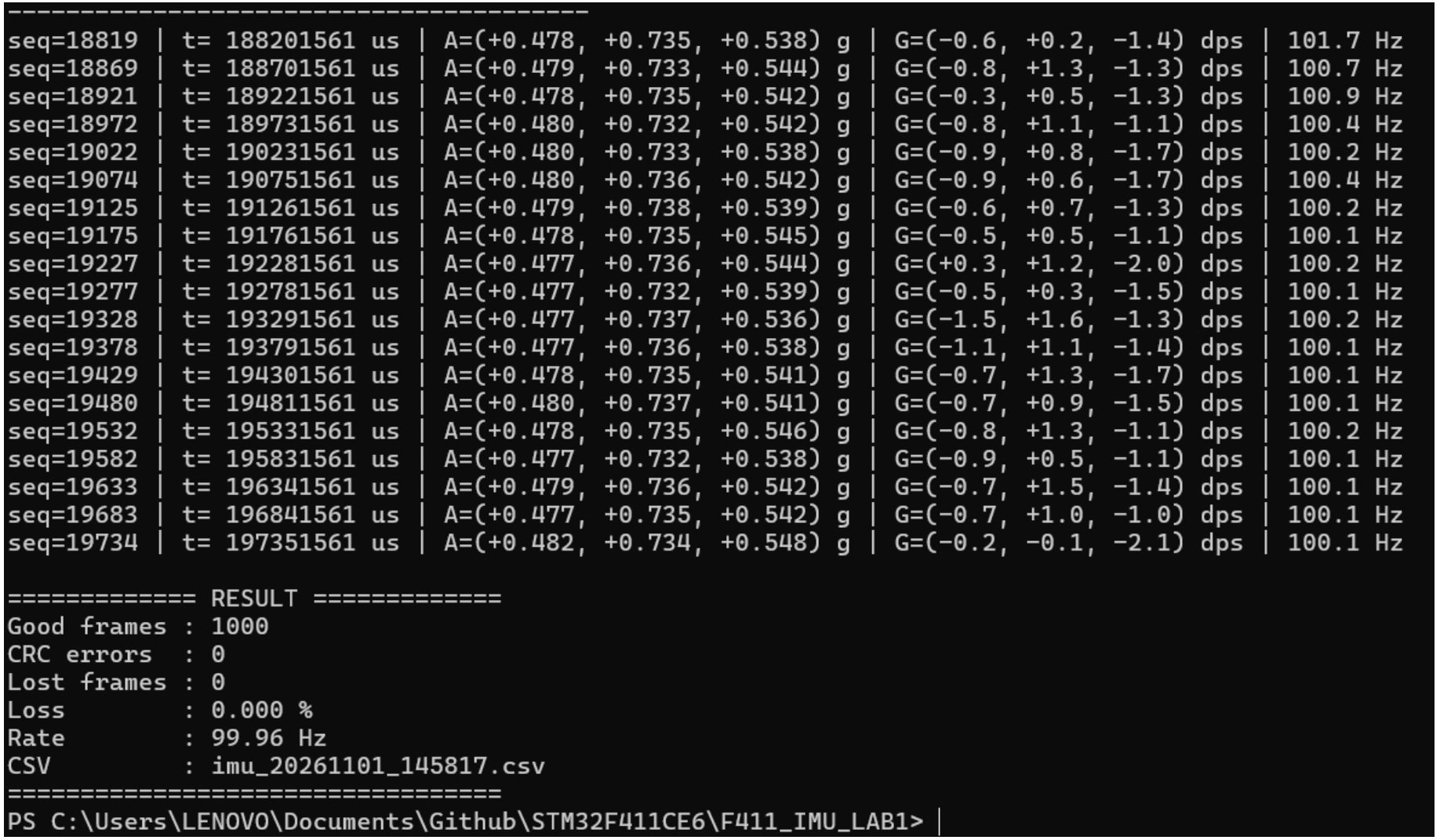


Original evidence: docs/evidence/08_com5_original.png

12. Python benchmark for 10 s

Windows PowerShell / Python

1000 frames; 0 CRC; 0 lost; 99.96 Hz. The CSV audit confirms 1000 rows in imu_20261101_145817.csv.

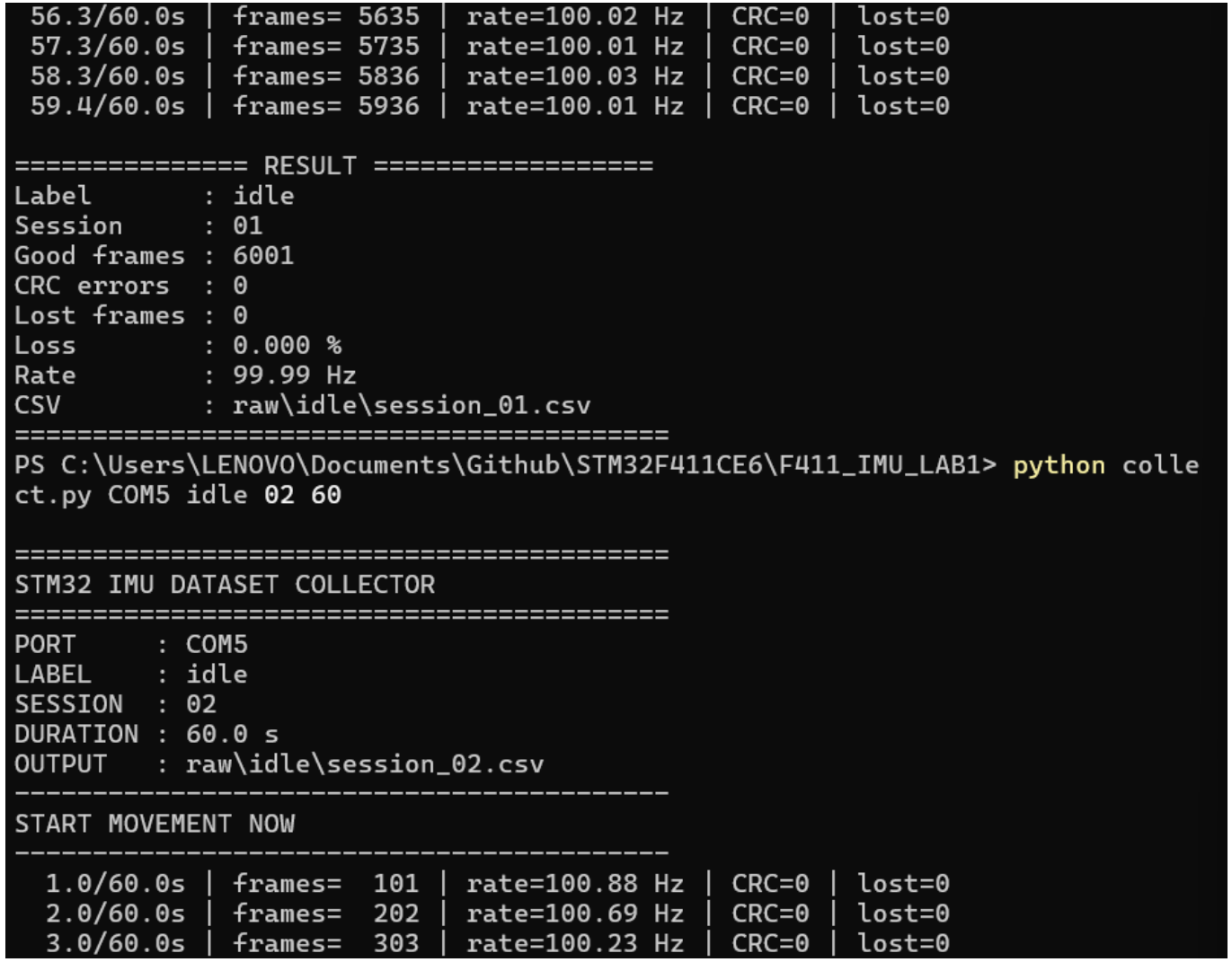


Original evidence: docs/evidence/09_benchmark_original.png

13. Collect by class and session

Windows PowerShell / Python

Top: idle 01 completed with 6001 frames at 99.99 Hz. Bottom: idle 02 starts a 60 s collection. Do not attribute the idle 01 result to idle 02.

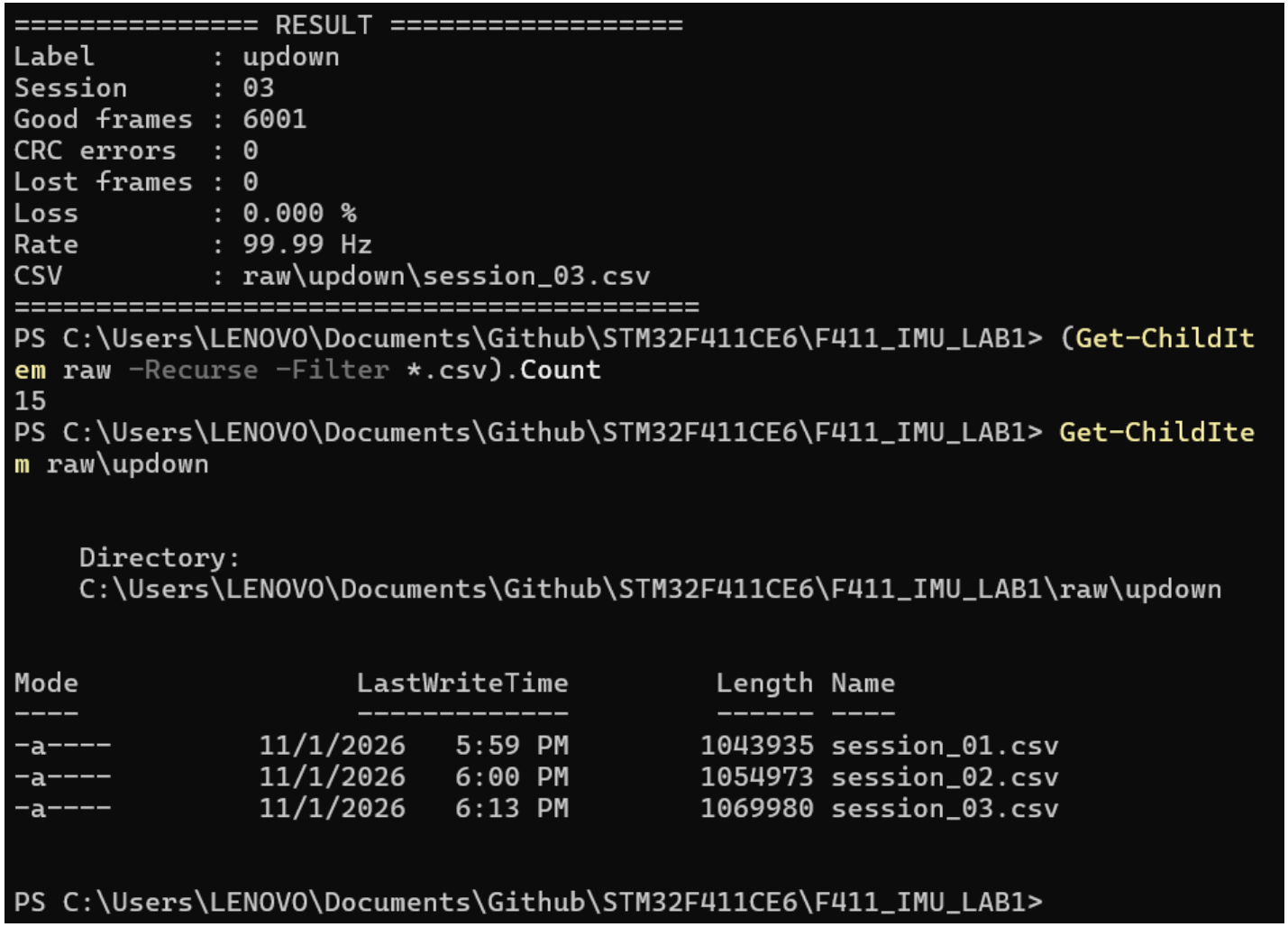


Original evidence: docs/evidence/10_collector_original.png

14. Complete all 15 CSV files

Windows PowerShell / Python

updown 03: 6001 frames; 0 CRC; 0 lost; 99.99 Hz. The count command returns 15; updown lists session_01/02/03. The file audit confirms 5 classes x 3 sessions.



Original evidence: docs/evidence/11_dataset_complete_original.png